

GREELEY-WELD COUNTY AP, CO, USA

WMO: 724768

Lat: 40.436N

Lon: 104.632W

Elev: 1432

StdP: 85.27

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.0	-17.6	-24.9	0.5	-21.5	-21.2	0.7	-16.2	14.6	1.2	12.8	0.7	1.2	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	35.9	17.1	33.8	16.9	32.3	16.8	19.7	29.4	19.0	29.0	18.3	28.4	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	14.5	22.8	16.1	13.7	22.1	15.0	12.7	21.9	62.7	29.4	60.3	29.1	57.7	28.6	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.7	8.7	-27.3	38.3	3.3	1.2	-29.7	39.2	-31.6	39.9	-33.4	40.6	-35.8	41.4
			-27.3	21.0	3.5	1.1	-29.8	21.7	-31.9	22.4	-33.9	23.0	-36.4	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.2	-1.2	4.5	8.6	13.9	20.0	23.3	21.8	17.1	9.3	2.6	-3.2		
	DBStd	10.44	6.07	6.20	5.18	4.53	4.41	3.61	2.31	2.38	4.02	4.74	5.27	5.96		
	HDD10.0	1696	408	314	177	77	16	1	0	0	4	67	224	409		
	HDD18.3	3595	666	547	428	291	146	24	1	4	68	279	473	668		
	CDD10.0	1524	0	1	7	36	138	302	414	364	216	46	2	0		
	CDD18.3	381	0	0	0	0	9	75	157	110	29	1	0	0		
	CDH23.3	5606	0	0	10	52	293	1161	1971	1403	636	79	1	0		
	CDH26.7	2616	0	0	1	4	93	543	1042	659	259	16	0	0		
(14) Wind	WSAvg	3.1	2.6	3.2	3.7	4.4	3.7	3.3	2.8	2.6	2.6	2.9	2.7	2.7		
(15) Precipitation	PrecAvg	341	8	9	18	42	57	41	42	43	30	28	13	9		
	PrecMax	462	22	24	91	173	154	98	93	144	104	84	41	32		
	PrecMin	191	0	0	0	3	16	2	7	2	1	7	2	2		
	PrecStd	70	7	6	23	33	31	26	23	30	23	18	10	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.0	24.7	27.4	32.5	37.3	37.8	36.1	34.4	28.7	22.3	16.0		
		MCWB	5.9	7.2	9.5	11.3	14.9	16.6	17.5	16.9	16.4	13.0	9.1	6.0		
	2%	DB	12.4	15.9	21.3	24.9	29.0	34.1	36.2	33.9	32.2	26.0	18.7	12.2		
		MCWB	4.3	5.8	8.5	10.6	13.9	16.1	17.5	17.0	15.5	11.8	8.4	4.4		
	5%	DB	8.9	12.3	18.6	22.5	27.1	32.4	34.1	32.5	29.9	22.6	16.0	8.8		
		MCWB	2.7	4.2	7.7	9.8	13.4	15.9	17.5	17.1	14.9	10.8	7.1	2.6		
	10%	DB	6.2	8.8	15.9	19.2	23.8	29.9	32.5	30.9	27.4	19.1	12.3	6.0		
		MCWB	1.1	2.5	6.3	8.8	12.6	15.7	17.6	17.2	14.7	9.6	5.3	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.8	10.3	12.7	17.3	19.8	20.8	20.4	18.6	14.0	9.9	6.5		
		MCDB	15.1	17.8	22.7	23.2	26.9	29.7	29.9	29.9	29.1	25.3	20.4	15.1		
	2%	WB	4.6	6.0	9.0	11.4	15.6	18.4	20.0	19.3	17.2	12.6	8.5	4.4		
		MCDB	11.6	14.9	20.0	22.0	24.8	28.7	29.6	28.9	27.9	23.1	18.1	11.4		
	5%	WB	3.0	4.4	7.8	10.4	14.4	17.4	19.3	18.6	16.2	11.4	7.1	2.7		
		MCDB	8.7	11.8	17.9	20.8	23.7	28.2	29.3	28.3	26.6	21.1	15.6	8.5		
	10%	WB	1.3	2.6	6.4	9.3	13.4	16.5	18.7	17.9	15.2	10.1	5.3	0.9		
		MCDB	5.9	8.4	15.0	18.8	22.4	27.6	29.0	27.6	24.7	18.6	11.9	5.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	15.2	16.9	16.0	15.4	16.9	17.7	17.4	17.7	16.7	15.9	14.8		
		MCDBR	19.7	20.8	23.1	22.8	20.8	21.4	21.1	20.8	22.5	22.9	21.5	19.3		
	5% WB	MCWBR	12.8	12.4	12.2	11.0	8.4	7.4	6.7	7.2	8.7	11.2	12.4	12.6		
		MCWBR	18.7	19.5	21.8	20.3	17.9	18.3	17.5	17.7	19.0	20.2	20.9	18.3		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.252	0.273	0.304	0.327	0.333	0.361	0.352	0.315	0.275	0.257	0.246		
		taud	2.599	2.583	2.533	2.462	2.454	2.460	2.422	2.437	2.520	2.613	2.577	2.579		
	Ebn at Noon		954	988	991	971	948	939	908	909	929	945	923	920		
		Edh at Noon	66	78	93	107	111	110	114	109	93	76	67	63		
(44) All-Sky Solar Radiation	RadAvg		2.42	3.36	4.64	5.55	6.24	7.26	7.00	6.17	5.21	3.71	2.65	2.08		
	RadStd		0.11	0.21	0.33	0.38	0.51	0.36	0.28	0.18	0.26	0.25	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.49	+0.96	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page